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sensemaking. The teacher projects (or writes on the board) the following table, left column filled in, right columns blank:

Written in full	Index form	Base	Exponent	Value
4,397,073 ≈ 4,400,000	4.4×10^6	10	6	1,000,000
1,155,574 ≈ 1,200,000	?	?	?	?
3,500,000 (maize, KNBS)	?	?	?	?
$2 \times 2 \times 2 \times 2 \times 2$	?	2	5	?
$3 \times$	?	3	4	?

have 3 minutes — working alone — to fill in every blank cell. Use your expand work from Phase 2.'

2. After 3 minutes, cold-call for each blank row — always WAIT 10 seconds. For the Kisumu row: 'Opiyo, what index form did you write for 1.2 million? Read the whole expression.' Probe: 'Why 10^6 and not 10^5 — convince me.'

3. For the maize row, invite the pair from Phase 2 who wrote 35×10^5 : 'Fatuma, both 3.5×10^6 and 35×10^5 equal 3,500,000 — are they both correct index form? Discuss with your partner for 20 seconds.' After pairs discuss, clarify: 'Both are mathematically equal. Scientists and the Kenya Meteorological Department use a convention: the leading number should be at least 1 but less than 10. We call this standard form. Today we are building that habit.'

4. Return to Phase 1 predictions on the board. Circle the off-by-one errors. Ask: 'Who changed their prediction? Tell me what you now know that you did not know 20 minutes ago.' Cold-call 2 learners.

5. Say: 'Write ONE sentence — your rule — that connects the exponent to the number of zeros. Do it now, 1 minute.' After writing, cold-call: 'Njeri, read your rule.' If incomplete, say: 'Good start — can anyone add to Njeri's rule?' Build the class consensus rule on the board.

6. Bridge to driving question: 'We can now READ and WRITE index expressions. But look at $10^6 \times 10^7$ — if the KNBS wanted to multiply Kisumu's population in millions by the national population in tens of millions, what would the answer look like in index form? PREDICT

Argumentation. The structured table constrains the cognitive load so learners focus on the relationship between the written number, the base, and the exponent — not on arithmetic. Requiring learners to write a rule 'in their own words' is a metacognitive consolidation move (self-explanation effect) that forces elaboration beyond procedural imitation.

learner's table row for Kisumu should read ' 1.2×10^6 , base = 10, exponent = 6, value = 1,000,000'. (b) The one-sentence rule should contain both the word 'exponent' (or 'power') and a reference to counting (digits, zeros, or multiplications). Vague rules such as 'the small number means big' are flagged — the teacher writes them on the board and asks the class to make them more precise before moving on.